

Default encryption at rest

This content was last updated in May 2024 and represents the status quo as of the time that it was written. Google's security policies and systems may change going forward, as we continually improve protection for our customers.

At Google, our comprehensive security strategy includes encryption at rest, which helps to protect customer data from attackers. We encrypt all Google customer content at rest, without any action required by you, using one or more encryption mechanisms. This document describes our approach to default encryption at rest for Google infrastructure and Google Cloud, and how we use it to keep customer content more secure.

This document is for security architects and security teams who are currently using or considering Google. This document assumes a basic understanding of [encryption](https://wikipedia.org/wiki/Encryption) (<https://wikipedia.org/wiki/Encryption>) and [cryptographic primitives](https://wikipedia.org/wiki/Cryptographic_primitive) (https://wikipedia.org/wiki/Cryptographic_primitive). For more information on cryptography, see [Introduction to Modern Cryptography](https://www.cs.umd.edu/%7Ejkatz/imc.html). (<https://www.cs.umd.edu/%7Ejkatz/imc.html>).

Encryption at rest is encryption that is used to help protect data that is stored on a disk (including solid-state drives) or backup media. All data that is stored by Google is encrypted at the storage layer using the Advanced Encryption Standard (AES) algorithm, AES-256. We use a common cryptographic library, Tink, which includes our FIPS 140-2 validated module (named [BoringCrypto](/security/compliance/fips-140-2-validated) (/security/compliance/fips-140-2-validated)) to implement encryption consistently across Google Cloud.

We own and manage the keys used in default encryption at rest. If you use Google Cloud, Cloud Key Management Service lets you create your own encryption keys that you can use to add envelope encryption to your data. Using Cloud KMS, you can create, rotate, track, and delete keys. For more information, see [Cloud Key Management Service deep dive](/docs/security/key-management-deep-dive) (/docs/security/key-management-deep-dive).

Keys in Google Cloud

The following table describes the different properties of keys in Google Cloud.

Type of key	Cloud KMS Autokey	Cloud KMS customer-managed (manual)	Google-owned and Google-managed encryption key (Google default encryption)
Can view key metadata	Yes	Yes	No

Ownership of keys ¹ (#footnote1)	Customer	Customer	Google
Can manage ² (#footnote2) and control ³ (#footnote3) keys	Key creation and assignment is automated. Customer manual control is fully supported.	Customer, manual control only	Google
Supports regulatory requirements for customer-managed keys	Yes	Yes	No
Key sharing	Unique to a customer	Unique to a customer	Data from multiple customers is typically protected by shared key encryption keys (KEKs).
Control of key rotation	Yes	Yes	<u>No</u> (/docs/security/encryption/default- encryption#kek_management)
<u>CMEK organization policies</u> (/kms/docs/cmek-org- policy)	Yes	Yes	No
<u>Log administrative and data access to encryption keys</u> (/kms/docs/audit- logging)	Yes	Yes	No
Logical data separation through encryption	Yes	Yes	<u>No</u> (/docs/security/encryption/default- encryption#hardware)
Pricing	<u>Varies</u> (/kms/pricing)	<u>Varies</u> (/kms/pricing)	Free

¹ The owner of the key indicates who holds the rights to the key. Keys that you own have tightly restricted access or no access by Google.

² Management of keys includes the following tasks:

- Create keys.
- Choose the protection level of the keys.

- Assign authority for management of the keys.
- Control access to keys.
- Control usage of keys.
- Set and modify the rotation period of keys, or trigger a rotation of keys.
- Change key status.
- Destroy key versions.

³ Control of keys means setting controls on the kind of keys and how the keys are used, detecting variance, and planning corrective action if needed. You can control your keys, but delegate management of the keys to a third party.

How encryption at rest helps to secure data

Encryption at rest is one piece of a broader security strategy. Encryption has the following benefits:

- Helps to ensure that if data falls into an attacker's hands, the attacker cannot read the data without also having access to the encryption keys. Even if attackers obtain the storage devices that contain customer data, they won't be able to understand or decrypt it.
- Helps to reduce the surface of attack by cutting out the lower layers of the hardware and software stack.
- Acts as a chokepoint because centrally managed encryption keys create a single place where access to data is enforced and can be audited.
- Helps to reduce the attack surface because instead of having to protect all data, businesses can focus their protection strategies on the encryption keys.
- Provides an important privacy mechanism for our customers. When data is encrypted at rest, it limits the access that systems and engineers have to the data

What is customer data?

As defined in the [Google Cloud Terms of Service](https://cloud.google.com/terms/) (<https://cloud.google.com/terms/>), *customer data* is data that customers or end users provide to Google through the services under their account.